

Technical Guide - Deployment & Operations

1. Deployment Process

The deployment process follows these steps:

1. Create a feature branch from develop.
2. Implement changes and write unit tests (minimum 80% coverage).
3. Submit a pull request with at least 2 reviewer approvals.
4. CI pipeline runs: lint, test, security scan, build.
5. Deploy to staging environment for QA validation.
6. After QA sign-off, create release PR to main.
7. Production deployment is automated via ArgoCD.
8. Post-deployment health checks run automatically.

Deployment windows: Tuesday-Thursday, 10 AM - 2 PM EST.

Emergency deployments require VP Engineering approval.

2. Incident Management

P1 (Critical): Full service outage. Response time: 15 minutes.

P2 (High): Major feature degraded. Response time: 1 hour.

P3 (Medium): Minor feature impact. Response time: 4 hours.

P4 (Low): Cosmetic/minor issue. Response time: 24 hours.

On-call rotation is managed through PagerDuty. All incidents must have a post-mortem within 48 hours.

Blameless culture: focus on system improvements, not individual fault.

3. Compliance Guidelines

All code must pass the following compliance checks:

- OWASP Top 10 security scan (zero critical findings)
- Data classification labeling on all database columns
- PII detection scan (no unmasked PII in logs)
- License compliance scan (no GPL dependencies in proprietary code)
- Accessibility audit (WCAG 2.1 AA compliance for web interfaces)

Quarterly compliance audits are conducted by the security team. Non-compliant services face access restrictions until remediated.